(20) $\int e^{-7x} \sin 5x \, dx =$

(1) $\frac{e^{-7x}}{74}[-7\sin 5x - 5\cos 5x] + c$

(2) $\frac{e^{-7x}}{74}[7\sin 5x + 5\cos 5x] + c$

(3) $\frac{e^{-7x}}{74}[7\sin 5x - 5\cos 5x] + c$

(4) $\frac{e^{-7x}}{74}[-7\sin 5x + 5\cos 5x] + c$

(21) $\int x^2 e^{\frac{x}{2}} dx =$

(1) $x^2 e^{\frac{x}{2}} - 4xe^{\frac{x}{2}} - 8e^{\frac{x}{2}} + c$

(2) $2x^2 e^{\frac{x}{2}} - 8xe^{\frac{x}{2}} - 16e^{\frac{x}{2}} + c$

(3) $2x^2 e^{\frac{x}{2}} - 8xe^{\frac{x}{2}} + 16e^{\frac{x}{2}} + c$

(4) $x^2 \frac{e^{\frac{x}{2}}}{2} - \frac{xe^{\frac{x}{2}}}{4} + \frac{e^{\frac{x}{2}}}{8} + c$

(22) $\int \frac{x+2}{\sqrt{x^2-1}} dx =$

(1) $\sqrt{x^2-1} - 2\log|x + \sqrt{x^2-1}| + c$

(2) $\sin^{-1} x - 2\log|x + \sqrt{x^2-1}| + c$

(3) $2\log|x + \sqrt{x^2-1}| - \sin^{-1} x + c$

(4) $\sqrt{x^2-1} + 2\log|x + \sqrt{x^2-1}| + c$

(23) $\int \frac{1}{x\sqrt{(\log x)^2 - 5}} dx =$

(1) $\log|x + \sqrt{x^2-5}| + c$

(2) $\log|\log x + \sqrt{\log x - 5}| + c$

(3) $\log|\log x + \sqrt{(\log x)^2 - 5}| + c$

(4) $\log|\log x - \sqrt{(\log x)^2 - 5}| + c$

(24) $\int \sin\sqrt{x} dx =$

(1) $2\left(-\sqrt{x}\cos\sqrt{x} + \sin\sqrt{x}\right) + c$

(2) $2\left(-\sqrt{x}\cos\sqrt{x} - \sin\sqrt{x}\right) + c$

(3) $2\left(-\sqrt{x}\sin\sqrt{x} - \cos\sqrt{x}\right) + c$

(4) $2\left(-\sqrt{x}\sin\sqrt{x} + \cos\sqrt{x}\right) + c$

(25) $\int e^{\sqrt{x}} dx =$

(1) $2\sqrt{x}(1 - e^{\sqrt{x}}) + c$

(2) $2\sqrt{x}(e^{\sqrt{x}} - 1) + c$

(3) $2e^{\sqrt{x}}(1 - \sqrt{x}) + c$

(4) $2e^{\sqrt{x}}(\sqrt{x} - 1) + c$